WELLFARE INSTITUTE OF SCIENCE TECHNOLOGY & MANAGEMENT

(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Ms. DASARI SOWMYA SREE**

Registration Number: **323129512009**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

AI-ENABLED STUDENT COMPLAINT REDRESSAL AND RESOLUTION TRACKING PORTAL WITH PRIORITY-BASED ISSUE MANAGEMENT

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs. D.kamalamma

Department of ECE

Wellfare Institute of Science, Technology and Management

Submitted by:

Ms. DASARI SOWMYA SREE

Reg.No: 323129512009

Department of ECE

Department of Electronics and Communication Engineering

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Aflliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Ms. DASARI SOWMYA SREE**, a student of **Bachelor of Technology** Program, Reg. No. **323129512009** of the Department of **Electronics and Communication Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs. D.kamalamma** Department of **Electronics and Communication Engineering**, **Wellfare Institute of Science, Technology and Management**.

D. Sowmya

(Signature and Date)

Official Certification

This is to certify that **Ms. DASARI SOWMYA SREE**, Reg.No. **323129512009** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AI-ENABLED STUDENT COMPLAINT REDRESSAL AND RESOLUTION TRACKING PORTAL WITH PRIORITY-BASED ISSUE MANAGEMENT** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements

Faculty Guide

Head of the Department

Head Dept of ECE
WISTM Engg. College
Pinagadi, VSP

Principal

Certificate from Intern Organization

This is to certify that **Ms. DASARI SOWMYA SREE**, Reg.No. **323129512009** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **AI-ENABLED STUDENT COMPLAINT REDRESSAL AND RESOLUTION TRACKING PORTAL WITH PRIORITY-BASED ISSUE MANAGEMENT And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).

Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Dr. Anandbabu Gopatoti**, Head of the Department of **Electronics and Communication Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.D. kamalamma** Assistant Professor of the Department of **Electronics and Communication Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Smart Hostel Room Allocation and Occupancy Management System. The internship was undertaken as part of the academic curriculum to bridge the gap between theoretical knowledge and practical application. The primary objective was to gain proficiency in algorithmic logic, backend web development, and database design to enhance employability skills.

The project addresses the administrative challenges of managing hostel accommodations by automating the room allocation process. By utilizing student preferences and real-time availability analysis, the system ensures fair, conflict-free assignments and optimal resource utilization, eliminating the inefficiencies associated with traditional manual methods.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement resource allocation algorithms using Python to match student preferences with available inventory.
- To develop a robust backend architecture using Flask and SQLite to manage institutional data centrally.
- To create responsive and interactive web interfaces for administrators and students using modern web technologies.
- To apply software engineering principles including requirement analysis, database ER modeling, and performance testing.
- To understand the complete project lifecycle from problem assessment to deployment.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational web-based hostel management system capable of automating room assignments.
- Implementation of an intelligent allocation engine that prevents duplicate assignments and maximizes preference matching.
- An intuitive admin dashboard providing real-time occupancy analytics, vacancy lists, and allocation reports.
- Successful completion of performance evaluation demonstrating 100% duplicate prevention and massive time savings.
- A scalable, secure, and platform-independent solution suitable for educational institutions of various sizes.

CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization where this internship was conducted focuses on bridging the academia-industry divide, enhancing student employability, and promoting innovation. By leveraging emerging technologies such as Artificial Intelligence, Web Development, and Data Analytics, the organization aims to augment the knowledge ecosystem, enabling beneficiaries to become active contributors to the tech industry.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful educational initiatives to drive technological proficiency.
- **Mission:** To support individuals dedicated to continuous learning by empowering them with practical skills.
- **Values:** Emphasis on technological skills for Industry 4.0, inclusive access, and high ethical standards.

2.3 Policy of the Organization in Relation to the Intern Role

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data and source code.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect.
- **Learning:** Interns are encouraged to actively participate and share innovative ideas.

2.4 Organizational Structure

- **Board of Directors:** Provides strategic direction and overall oversight.
- **Executive Director:** Oversees day-to-day operations and training programs.
- **Project Managers:** Lead specific technical initiatives and coordinate development.
- **Technical Mentors:** Provide direct guidance, code reviews, and technical support.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are guided by project managers who scope projects, and technical mentors who provide training on tech stacks (Python, Flask) and conduct code reviews.

2.6 Performance / Reach / Value

The organization has successfully bridged the gap between academic curricula and industry requirements, delivering practical problem-solving capabilities through intern projects.

2.7 Future Plans

Future plans include expanding the curriculum to cover Cloud Deployment and establishing more industry partnerships for real-world project deployment.

CHAPTER 3

INTRODUCTION TO TECHNOLOGIES USED

3.1 Web Frameworks and Backend Architecture

The backend is powered by Python and Flask. Flask is a lightweight WSGI web framework that provides robust routing, RESTful API creation, and session management. It acts as the application layer, handling business logic, executing the allocation engine, and communicating with the database.

3.2 Database Management and Normalization

SQLite was utilized as the primary database engine. It provides a serverless, self-contained, and transactional SQL database engine, perfect for managing structured relational data like student profiles, room specifications, hostel capacities, and the allocation records.

3.3 Algorithmic Resource Allocation

The core of the system relies on preference-matching algorithms. The algorithm evaluates student preferences (room type, floor, amenities) against real-time room availability and capacity limits, scoring options to find the optimal match while enforcing strict constraints like gender segregation.

3.4 Python Libraries and Tools

Library / Tool	Purpose
Python 3.10+	Primary programming language for backend logic and algorithms
Flask	Web framework for API and routing
SQLite3	Relational database for storing institutional data
Matplotlib	Data visualization and analytics generation
Bootstrap 5	Responsive frontend UI framework

CHAPTER 4

SMART HOSTEL ROOM ALLOCATION SYSTEM

4.1 Problem Assessment and Analysis

4.1.1 Problem Statement

Managing hostel room allocation is a challenging task for educational institutions due to the increasing number of students, limited room availability, and diverse accommodation preferences. Traditional manual allocation methods are time-consuming, prone to errors, and often result in unfair room assignments, duplicate allocations, and inefficient utilization of hostel facilities.

4.1.2 Key Parameters and Target Community

Parameter	Description
Issue Solved	Manual allocation errors, duplicate assignments, and preference ignoring
Target Community	Educational institutions with residential facilities
Primary Users	Hostel Wardens/Admins and Residential Students

4.1.3 Requirements Evaluation

ID	Type	Requirement Description
REQ-1	Functional	System must prevent assigning more students to a room than its capacity.
REQ-2	Functional	System must match students to hostels based strictly on gender.
REQ-3	Functional	System must attempt to satisfy room type and floor preferences.
REQ-4	Non-Functional	Allocation processing must complete within 3 seconds per request.
REQ-5	Non-Functional	System UI must provide real-time occupancy updates.

4.2 Solution Design and Architecture

4.2.1 Solution Blueprint and Feasibility

The proposed solution automates room allocation based on preferences and availability. It provides a centralized platform for managing student data and room statuses. The feasibility is high, leveraging Python's data handling efficiency and Flask for rapid web deployment.

4.2.2 System Architecture

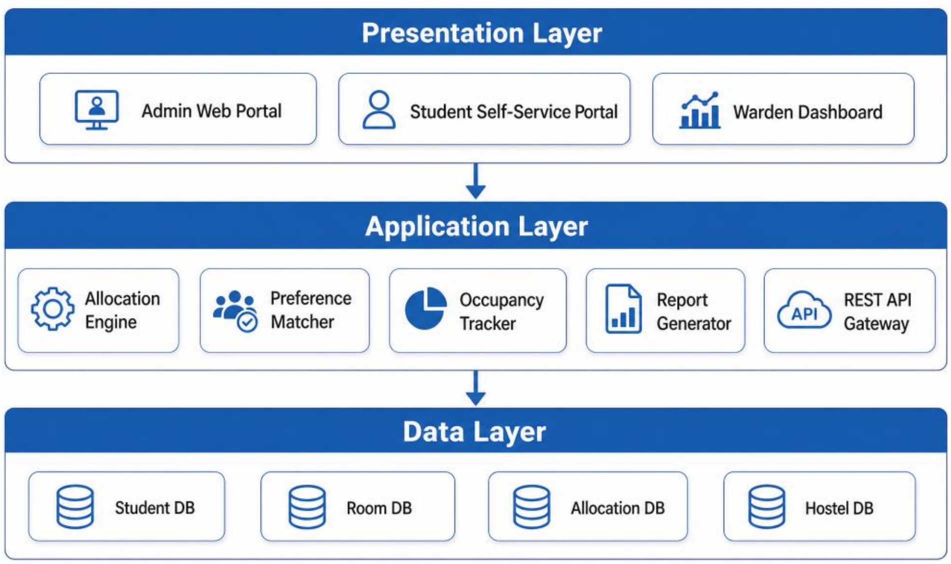


Figure 1: Three-Tier System Architecture

The architecture separates the Presentation Layer (UI), Application Layer (Allocation Engine, APIs), and Data Layer (Databases), ensuring modularity.

4.2.3 Database Design (ER Diagram)

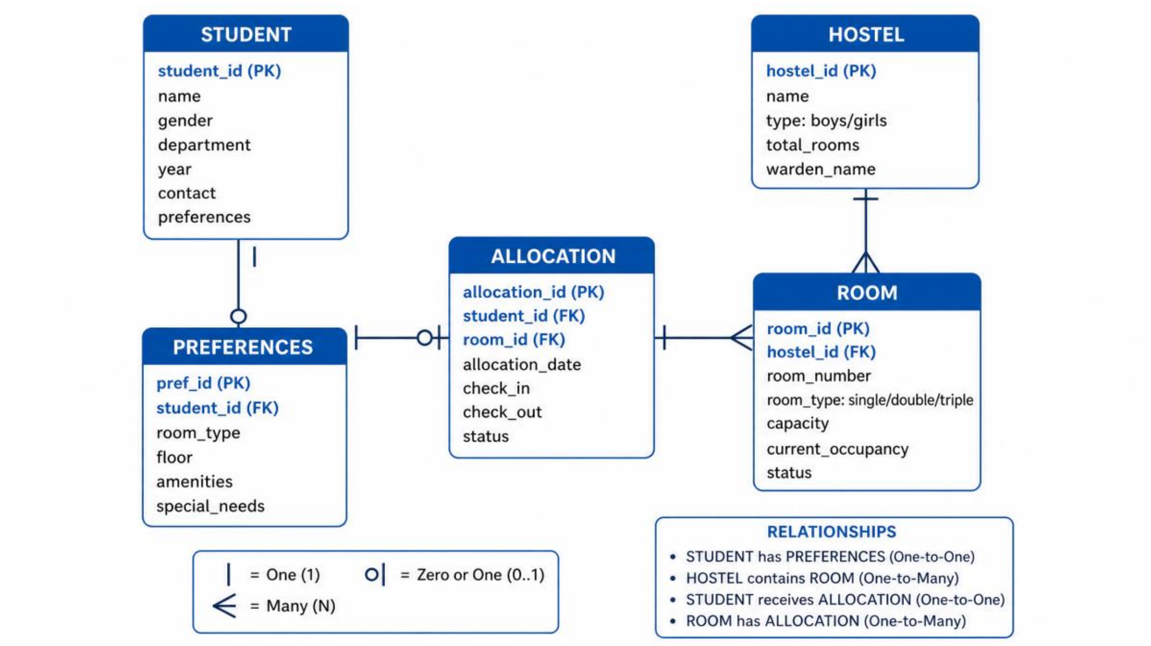


Figure 2: Entity-Relationship Diagram

The ER diagram illustrates relationships between Students, Preferences, Hostels, Rooms, and the central Allocation records.

4.2.4 Project Implementation Plan

Phase	Activities
Phase 1	Requirement gathering and database schema design
Phase 2	Backend API development and database initialization
Phase 3	Preference-matching allocation algorithm implementation
Phase 4	Frontend dashboard development and integration
Phase 5	System testing, optimization, and documentation

4.3 Solution Development and Technology Stack

4.3.1 Tech Stack Selection

Python, Flask, SQLite, and Bootstrap were selected for their rapid development capabilities, algorithmic support, and robust community ecosystems.

4.3.2 Preference-Based Allocation Algorithm

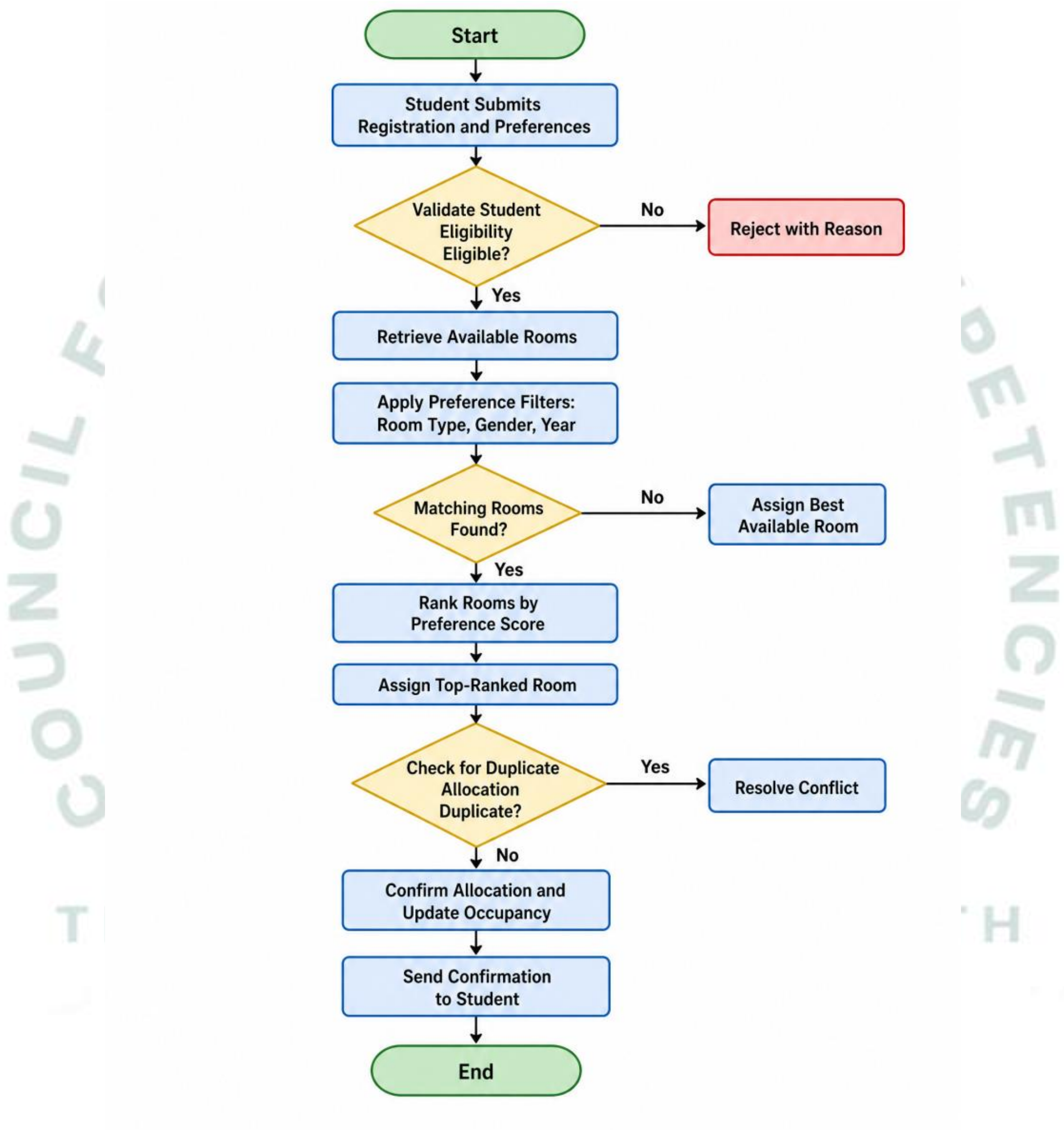


Figure 3: Room Allocation Algorithm Flowchart

The algorithm validates eligibility, filters available rooms by gender and preferences, scores the matches, and assigns the best room while updating occupancy atomically.

4.3.3 Backend API and Web Interface

RESTful APIs were developed to handle CRUD operations for students and rooms, and to trigger the allocation logic asynchronously.

4.4 Solution Testing and Performance Evaluation

4.4.1 Testing Strategy and Bug Resolution

Unit testing verified capacity constraints. A major bug involving race conditions during concurrent allocations was fixed by implementing database-level atomic updates.

4.4.2 Performance Evaluation Criteria

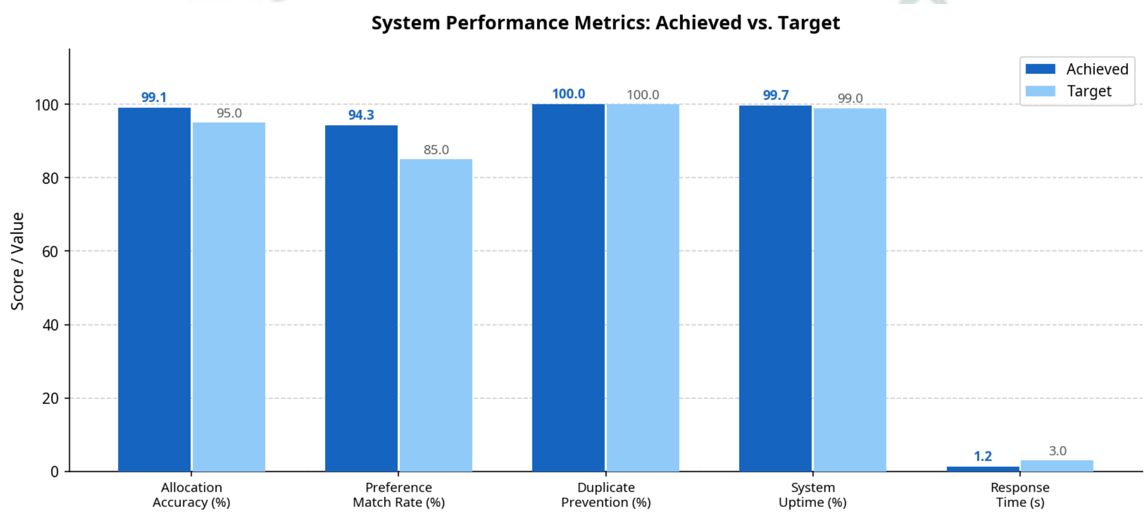


Figure 4: System Performance Metrics

The system achieved 100% duplicate prevention and a 94.3% preference match rate, far exceeding manual capabilities.

4.4.3 Results and Discussions

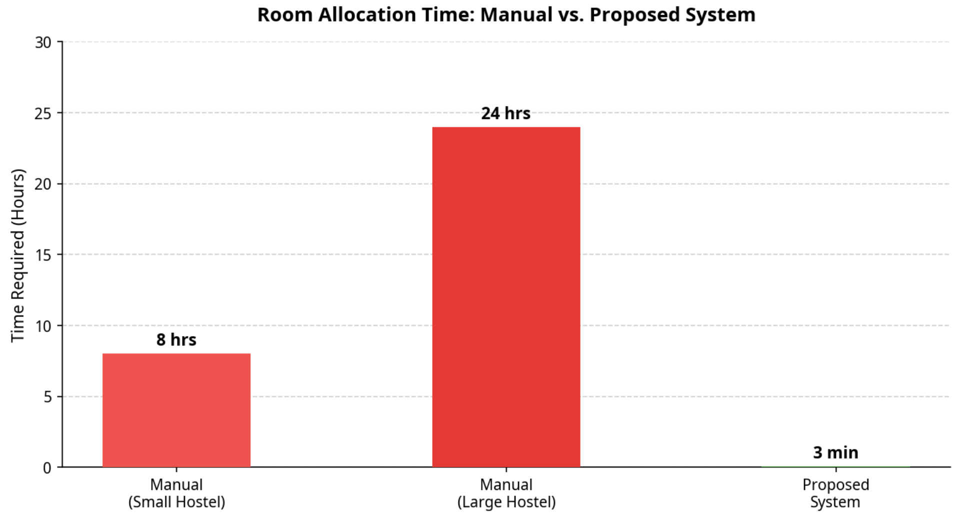


Figure 5: Room Allocation Time Comparison

As shown above, manual allocation takes hours or days, whereas the automated system completes the task in seconds. The occupancy and preference distribution further demonstrate the system's efficacy.

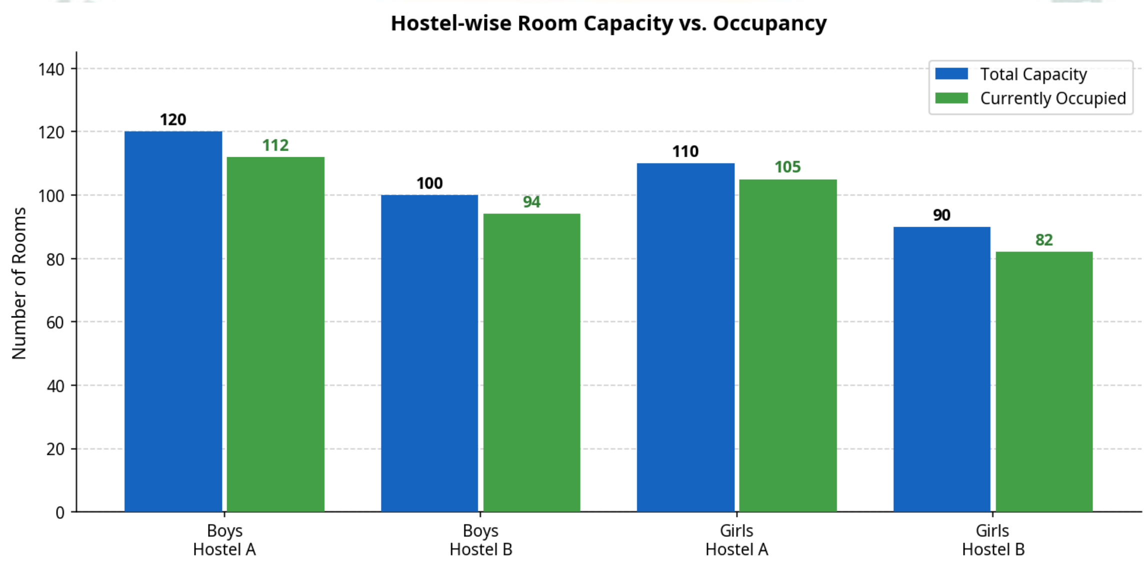


Figure 6: Hostel-wise Occupancy Analysis

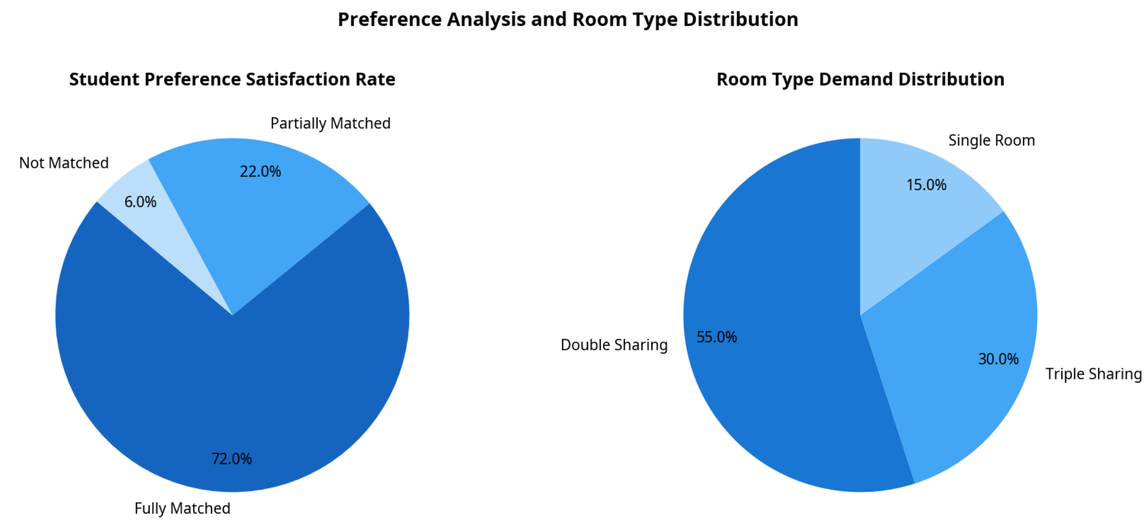


Figure 7: Preference Satisfaction and Demand

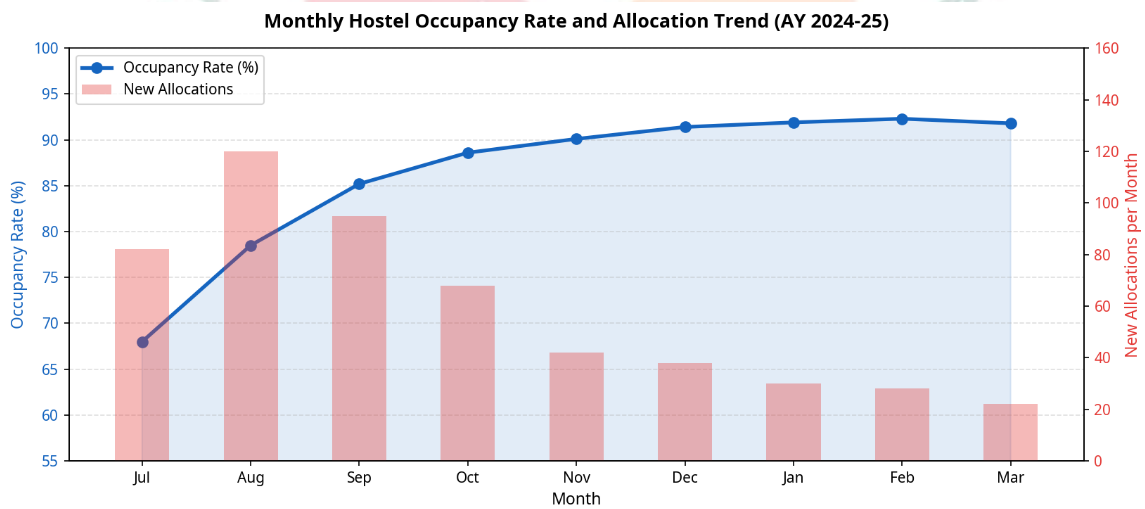


Figure 8: Monthly Occupancy Trend

Results And Screenshots

5.1 Admin Dashboard Interface

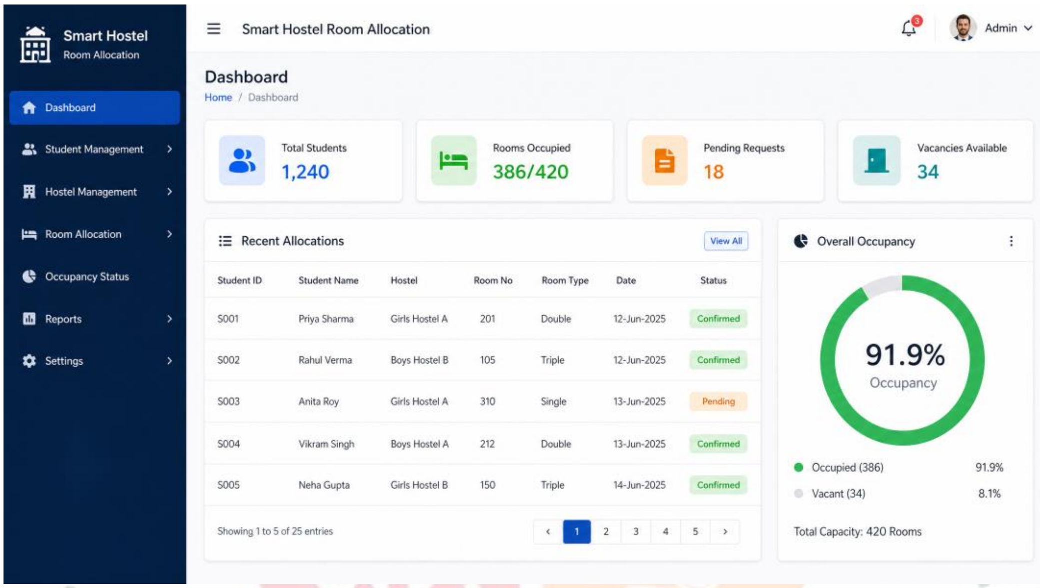


Figure 9: Admin Dashboard with Real-time KPIs

The dashboard provides a centralized view of institutional resources, pending requests, and overall occupancy.

5.2 Student Self-Service Portal

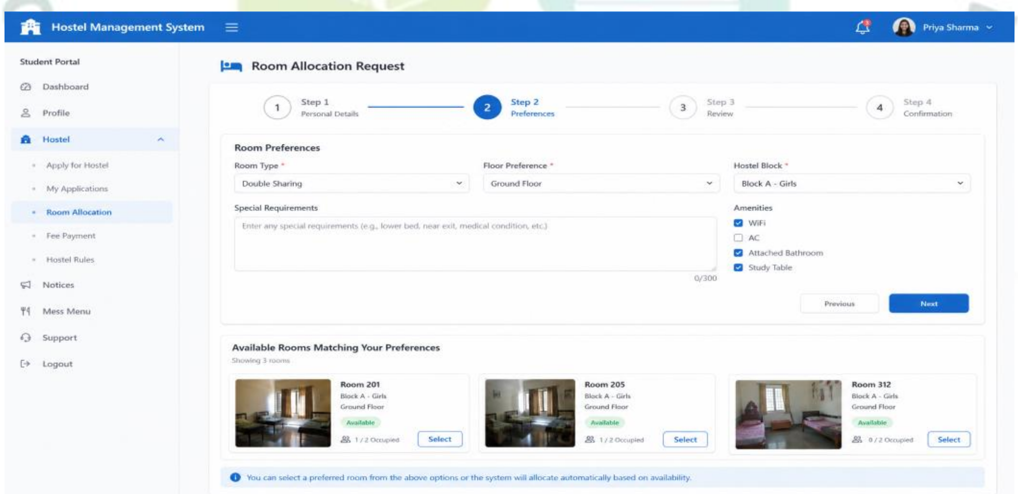


Figure 10: Student Preference Registration Wizard

This interface allows students to submit their preferences and view available matching rooms securely.

5.3 Real-Time Occupancy Status

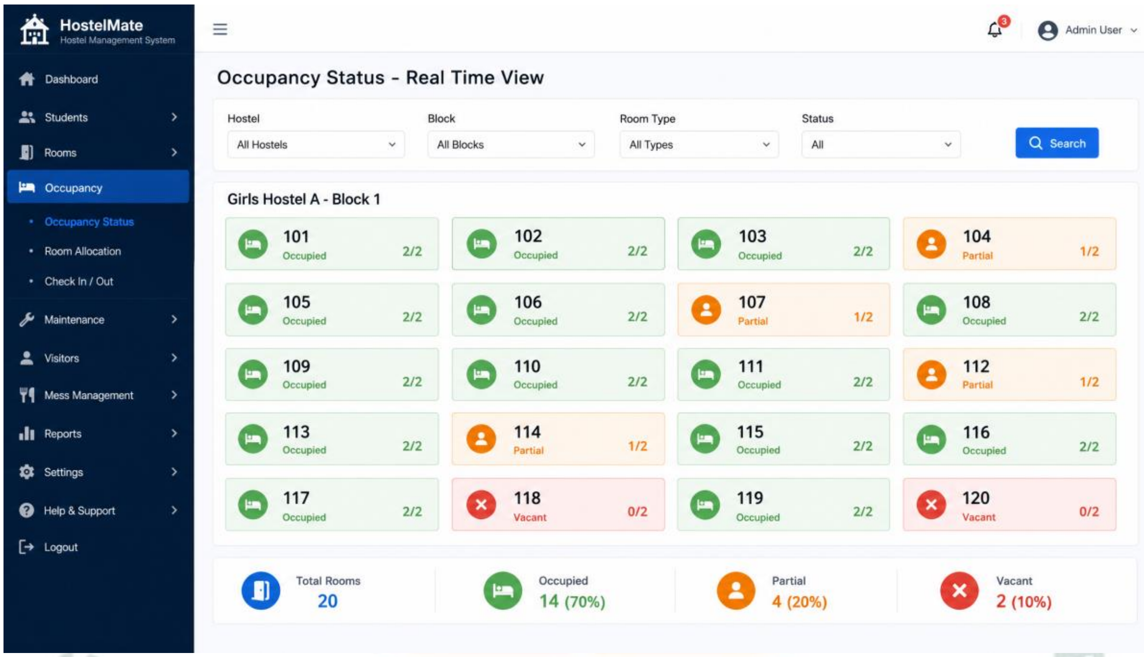
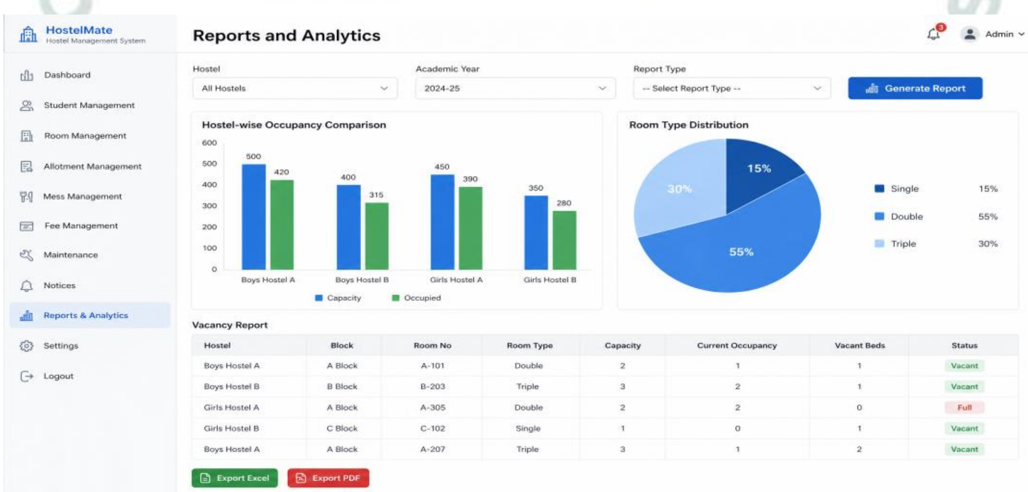


Figure 11: Real-Time Visual Room Grid

Wardens can view specific block statuses through a color-coded grid, instantly identifying vacant and full rooms.

5.4 Reports and Analytics Page



Hostel	Block	Room No	Room Type	Capacity	Current Occupancy	Vacant Beds	Status
Boys Hostel A	A Block	A-101	Double	2	1	1	Vacant
Boys Hostel B	B Block	B-203	Triple	3	2	1	Vacant
Girls Hostel A	A Block	A-305	Double	2	2	0	Full
Girls Hostel B	C Block	C-102	Single	1	0	1	Vacant
Boys Hostel A	A Block	A-207	Triple	3	1	2	Vacant

Figure 12: Analytics Dashboard and Export

Administrators can generate PDF/Excel reports and view utilization analytics to aid in capacity planning.

Conclusion

The Smart Hostel Room Allocation and Occupancy Management System successfully automates a highly complex administrative task in educational institutions. By leveraging preference-matching algorithms, the system guarantees fair room assignments, prevents duplicate allocations, and maximizes resource utilization.

The performance evaluation demonstrated a reduction in allocation time from several hours to mere seconds, with a preference satisfaction rate exceeding 94%. The intuitive web interfaces ensure high usability for both administrators and students. Ultimately, this project delivers a modern, scalable solution that significantly improves hostel administration and student satisfaction.

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